

```
#include <stdio.h>
```

```
#include <stdlib.h>
```

```
/* Definition of node */
```

```
struct node {
```

```
    int data;
```

```
    struct node *next;
```

```
};
```

```
struct node *head = NULL;
```

```
/* Function Prototypes */
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```
void createe();
```

```
void insert5AA();
```

```
void deletee();
```

```
void traverse();
```

```
/* Main Function */
```

```
int main() {
```

```
    int choice;
```

```
    while (1) {
```

```
        printf("\n--- Singly Linked List Menu ---\n");
```

```
        printf("1. Create\n");
```

```

printf("2. Insert5AA\n");

printf("3. Delete\n");

printf("4. Traverse\n");

printf("5. Exit\n");

printf("Enter your choice: ");

scanf("%d", &choice);


switch (choice) {

    case 1: create();

        break;

    case 2: insert();

        break;

    case 3: deletee();

        break;#include <stdio.h>
#include <stdlib.h>

struct node
{
    int data;
    struct node *next;
};

struct node *front = NULL, *rear = NULL;

/* Function declarations */
void enqueue();
void dequeue();
void peek();
void display();

```

```

int main()
{
    int choice;

    while(1)
    {
        printf("\n--- Queue Using Linked List ---\n");
        printf("1. Enqueue\n");
        printf("2. Dequeue\n");
        printf("3. Peek\n");
        printf("4. Display\n");
        printf("5. Exit\n");
        printf("Enter your choice: ");
        scanf("%d", &choice);

        switch(choice)
        {
            case 1: enqueue(); break;
            case 2: dequeue(); break;
            case 3: peek(); break;
            case 4: display(); break;
            case 5: exit(0);
            default: printf("Invalid choice\n");
        }
    }
}

/* Enqueue operation */
void enqueue()
{
    struct node *ptr;
    int value;

    ptr = (struct node*)malloc(sizeof(struct node));
    if(ptr == NULL)
    {
        printf("Queue Overflow\n");
        return;
    }

    printf("Enter element to insert: ");
    scanf("%d", &value);

    ptr->data = value;

```

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        ptr->next = NULL;

        if(rear == NULL)
        {
            front = rear = ptr;
        }
        else
        {
            rear->next = ptr;
            rear = ptr;
        }

        printf("Element inserted successfully\n");
    }

/* Dequeue operation */
void dequeue()
{
    struct node *temp;

    if(front == NULL)
    {
        printf("Queue Underflow\n");
        return;
    }

    temp = front;
    printf("Deleted element: %d\n", temp->data);
    front = front->next;

    if(front == NULL)
        rear = NULL;

    free(temp);
}

/* Peek operation */
void peek()
{
    if(front == NULL)
    {
        printf("Queue is empty\n");
        return;
    }

```

```
        printf("Front element: %d\n", front->data);
    }

/* Display operation */
void display()
{
    struct node *temp;

    if(front == NULL)
    {
        printf("Queue is empty\n");
        return;
    }

    temp = front;
    printf("Queue elements: ");
    while(temp != NULL)
    {
        printf("%d ", temp->data);
        temp = temp->next;
    }
    printf("\n");
}
```


